

DESIGN DOCUMENT

CURRENT FREE CASH FLOW YIELD

Status: Approved for V1.0.0 **Companions:** docs/current-fcf-yield-prd.md , docs/current-fcf-yield-techspec.md **Parent:** docs/architecture.md

1. Overview

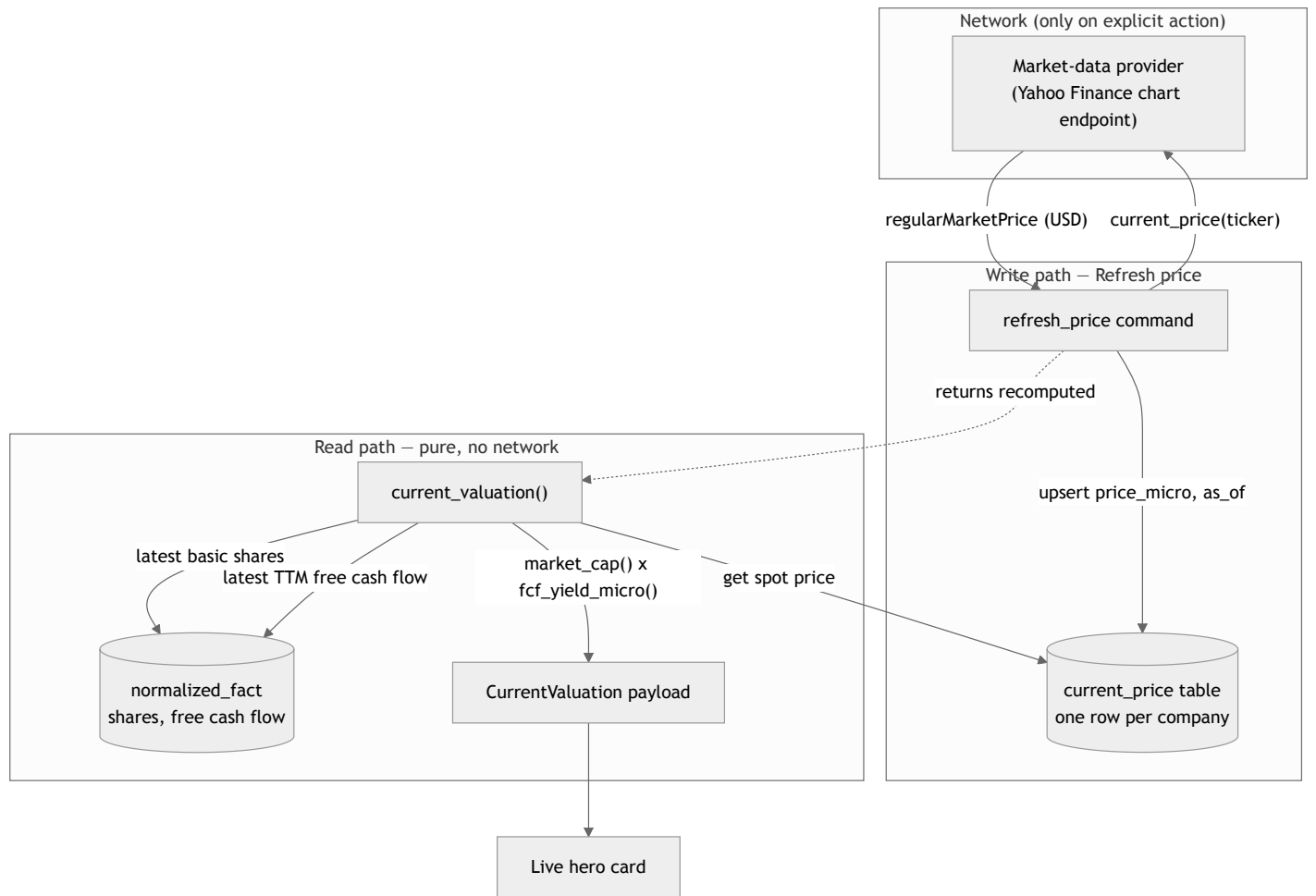
This feature adds a live valuation scalar — the **current free cash flow yield** — to a system whose entire metric layer is otherwise built on **read-time derivation from immutable normalized facts**. The design challenge is that a "current" figure depends on a value (today's stock price) that is *not* an SEC fact and changes continuously, while the architecture's correctness guarantees rest on never persisting a computed product.

The resolution is a clean split:

- The **one volatile input** that cannot be derived — the current spot price — is **persisted** in a dedicated single-row-per-company table, stamped with the time it was fetched.
- **Everything else** (market cap, the yield, the numerator selection) is **derived at read time** from that stored price plus the existing normalized facts, exactly like every other metric.

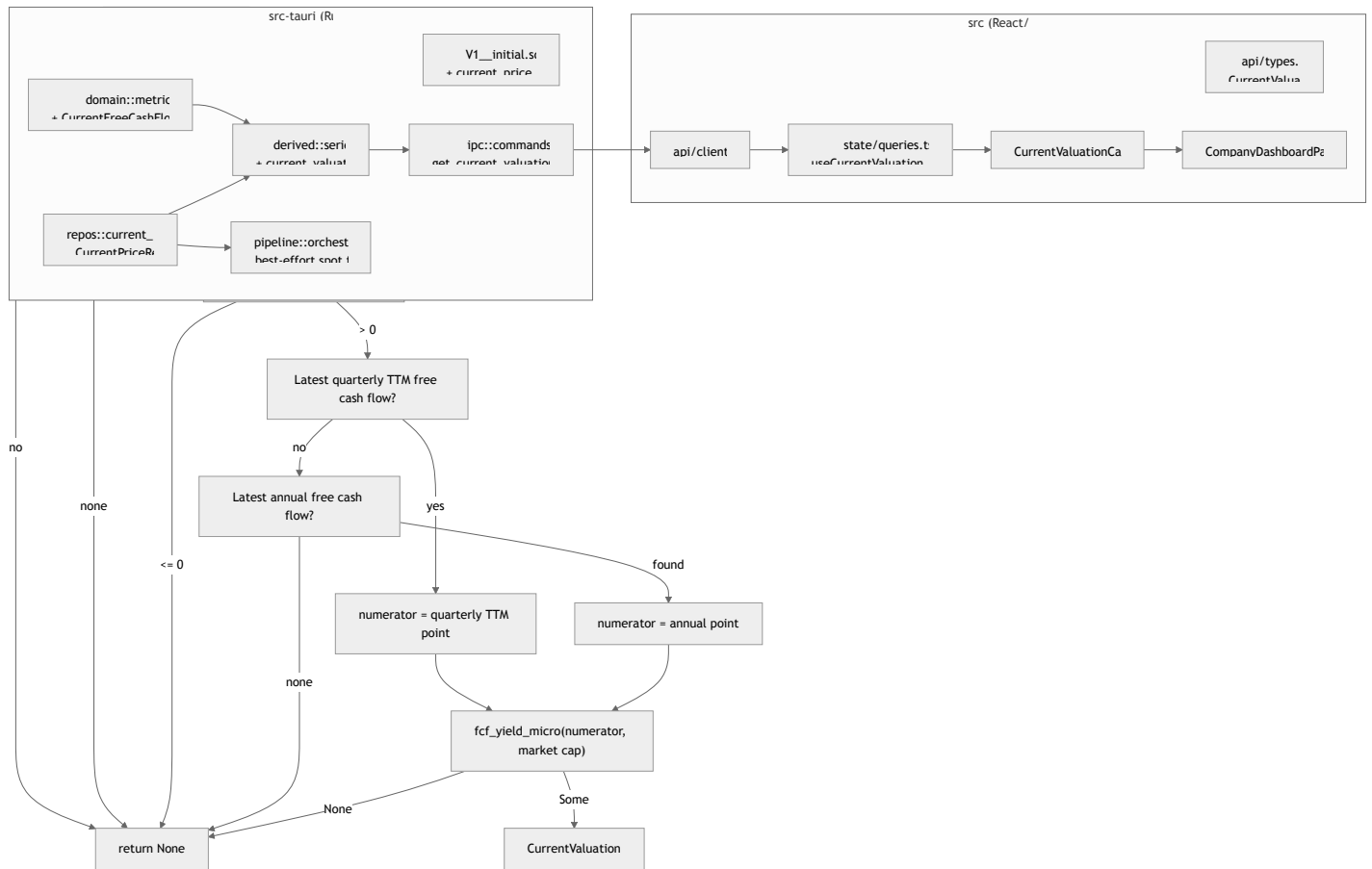
This keeps the read path offline and pure, confines all network I/O to an explicit user action, and reuses the existing pure formulas without introducing any new math.

2. Data flow



The full company-refresh pipeline also calls the provider for the spot price as a **best-effort** step: a price-fetch failure records a warning event and continues, never failing ingestion.

3. Computation



3.1 Numerator selection

The numerator is the most recent **quarterly trailing-twelve-month** free cash flow point, falling back to the most recent **annual** free cash flow point only when no quarterly trailing point exists. Quarterly trailing-twelve-month is the freshest reasonable measure of "current" free cash flow; the annual fallback ensures companies that report only annually still produce a figure.

3.2 Reused primitives

No new arithmetic is introduced. The design reuses two existing pure functions from the derived-metric layer:

- `market_cap(close_micro, shares)` — an `i128` product clamped to `i64`, where `close_micro` is `price × 1e6` and `shares` is a raw count.
- `fcf_yield_micro(free_cash_flow, market_cap)` — returns the decimal-ratio `× 1e6`, or `None` when market cap is non-positive. A negative free cash flow yields a valid negative ratio, which is preserved.

4. Key decisions & trade-offs

4.1 Persist the price; derive everything else

Decision. Store the spot price (the one input that is genuinely time-varying and not an SEC fact); derive market cap and yield at read time.

Alternatives considered.

- *Persist the computed market cap or the computed yield.* Rejected — it reintroduces exactly the desync risk the read-time architecture exists to eliminate: a later shares restatement would silently leave a stored product wrong.
- *Fetch the price live on every read.* Rejected — it breaks the offline-after-ingest guarantee, makes the dashboard slow and network-dependent, and couples a render to provider availability.

Trade-off. The displayed figure is only as fresh as the last Refresh. This is acceptable and is made honest by the freshness badge and timestamp stamps (see §5).

4.2 Honest freshness over implied liveness

Decision. The "LIVE" badge is conditional on the stored price being under 24 hours old; otherwise the badge reads "QUOTED · ". The price's `as_of` timestamp is always shown.

Rationale. Accuracy is a hard constraint, not a presentation preference. Labeling a three-day-old price as "LIVE" would be a form of inaccuracy. The card tells the truth about how current its inputs are.

4.3 Never fabricate

Decision. `current_valuation` returns `None` whenever any input is missing or market cap is non-positive; the UI renders a prompt, not a number.

Rationale. A plausible-but-wrong yield is worse than no yield. This mirrors the existing derived layer's `Option` discipline end to end through the IPC boundary to the UI's "No live quote yet" state.

4.4 A scalar, not a series

Decision. The current free cash flow yield is modeled as a single `CurrentValuation` payload returned by a dedicated command, **not** as a point in `get_metric_history`.

Rationale. It has a fundamentally different shape: one value "as of now," not a per-period series. The series dispatcher returns an empty vector for the `current_market_cap` and `current_free_cash_flow_yield` metric identifiers, with an explicit arm documenting that these are served by `current_valuation`. This keeps the chart/series code honest (it never invents a one-point "series") and gives the UI a purpose-built payload with all the freshness metadata it needs.

5. UI design — the live hero card

The card is the single attention-drawing element on the dashboard. Its visual language is deliberately divergent from the surrounding muted period-end tiles:

- **Ring + glow:** an accent-colored ring and a soft outer shadow elevate the card off the page.
- **Pulsing LIVE badge:** a small pill with a pulsing dot signals liveness when fresh; it degrades to a muted "QUOTED · " pill when stale (§4.2). The pulse is gated on `motion-safe`.
- **Sign-colored hero number:** green for a positive yield, red for negative; the sign is also legible in the number text itself, so color is never the sole signal.
- **Neutral delta chip:** the move versus the last period-end yield is shown in a neutral accent color with a directional arrow glyph — contextual information, not a value judgment.
- **Spelled-out math and freshness stamps:** the full computation and every input's vintage are visible, upholding the product's auditability principle.

State handling (loading / null / error / success) is enumerated in the PRD §6.3 and implemented in the card component.

6. Component & module impact

7. Release — version 1.0.0

This feature is the V1.0.0 capstone. The version bump touches:

Touchpoint	File
Rust crate version (drives <code>ping</code>)	<code>src-tauri/Cargo.toml</code>
Tauri bundle (drives DMG filename)	<code>src-tauri/tauri.conf.json</code>
npm package	<code>package.json</code> , <code>package-lock.json</code>
In-app footer string	<code>src/components/AppShell.tsx</code>
Mock <code>ping</code> version	<code>src/test-mock-auri.ts</code>
Installer references	<code>DEMO_README.md</code> , tracked DMG artifact

A fresh ad-hoc-signed DMG is built and the prior 0.1.0 artifact is replaced.